

Rethinking case attraction on Ancient Greek infinitive clauses

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- (1) a. συμβουλεύει τῷ Ξενοφῶντι ἐλθόντα εἰς Δελφοὺς ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
Ele aconselha Xenofonte ir a Delfos e interrogar o deus sobre a viagem. (Xen. *Anab.* 3.1.5)
b. ἀφῆκε μοι ἐλθόντι πρὸς ὑμᾶς λέγειν τᾶληθῆ.
Ele me permitiu vir frente a vós dizer a verdade. (Xen. *Hell.* 6.1.13)

Trabalhos mencionados: Lancelot (1658, 1709), Lakoff (1970), Andrews (1971), Quicoli (1982), Spyropoulos (2005), Tantalou (2003), Sevdali (2007, 2013)

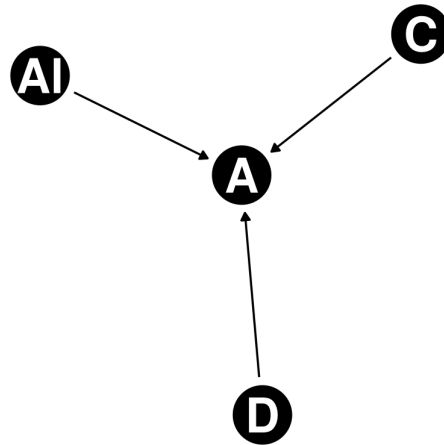
Fonte dos dados: Diorisis Corpus (Vatri e McGillivray 2018, 2020) with help of the project's querying tool, Diorisis Search (Vatri 2020-2022).

Corpus: Sentenças coletadas a partir das fontes literárias áticas e Heródoto.

Ferramentas: R (R Core Team 2013) com o os pacotes tidyverse (Wickham et al. 2019), dagitty (Textor et al. 2016), ggdiag (Barrett 2025) e rethinking (McElreath 2020), bem como o software Stan (Stan Development Team 2025a,b).

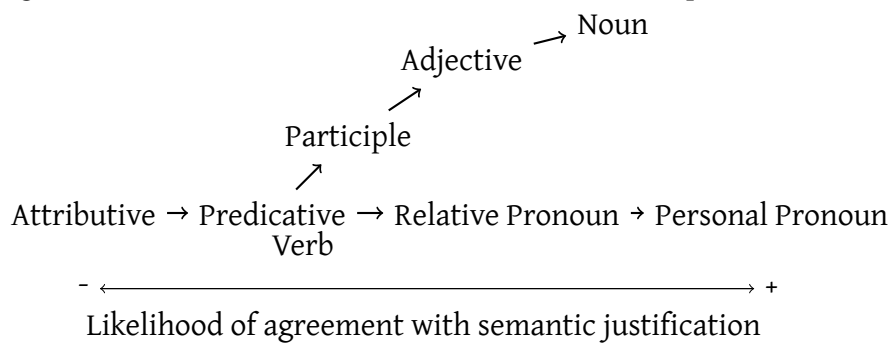
Modelo tipológico: Corbett (2006)

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Infinitivos copulares: ver Kühner e Gerth (1898)

(2) Agreement and Predicate Hierarchies (Corbett 2006, p. 233):



Verbos impessoais: ver Humbert (1945)

Verbos com semântica modal: verbos como ἔξαρκεῖ e ἔξεστί tem maior probabilidade de apresentar atração de caso (Geraldes 2020, 2025).

(3) ἄλλ' ἔξαρκῇ σοι τὰ ἔργα αὐτῶν ὁρῶντι σέβεσθαι καὶ τιμᾶν τοὺς θεοὺς.
Mas é suficiente a ti observar as ações e respeitar e honrar os deuses. (Xen. Mem. 4.3.13)

Autoria e dialeto: Geraldes (2020, 2025) mostra que o uso de verbos com semântica modal são mais frequentes em diálogos filosóficos que textos historiográficos.

Ordem de palavra e distância: maiores distâncias diminuem a probabilidade de atração de caso. Alvos que antecedem o controlador parecem exigir atração de caso como no exemplo (4).

(4) ὥς ἀγαθοῖς τε ὑμῖν προσήκει εἶναι.
Que vos seja dado ser boa gente. (Xen. Anab. 3.2.11)

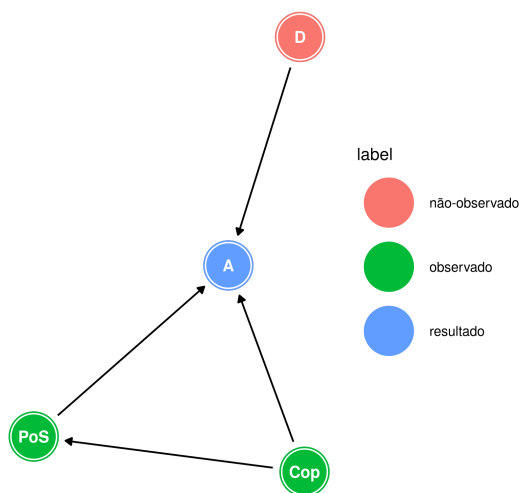


Figura 1: DAG with Attraction, **Domain**, **Part of Speech** and **Copula**

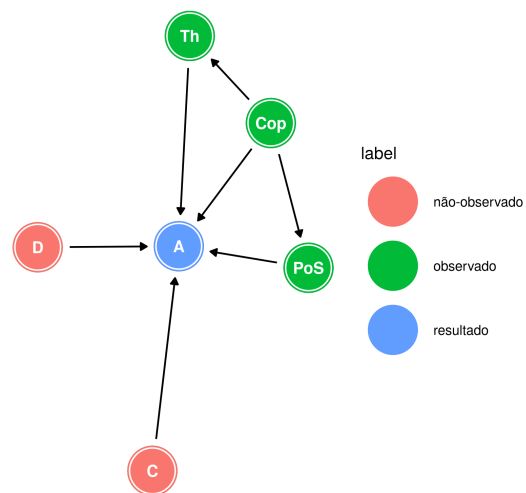


Figura 2: DAG with Attraction, **Domain**, **Controller Part of Speech**, **Copula** and **Thematic** role.

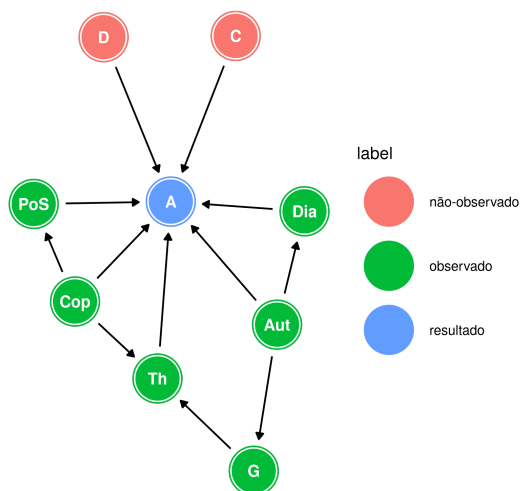


Figura 3: DAG with Attraction, **Domain**, **Controller Part of Speech**, **Copula**, **Thematic** role, **Authorship**, **Genre**, and **Dialect**.

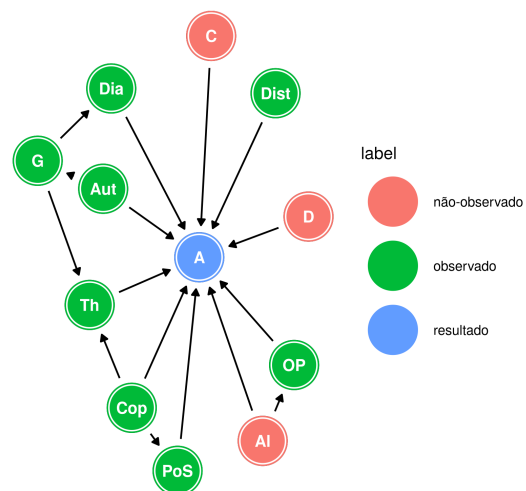
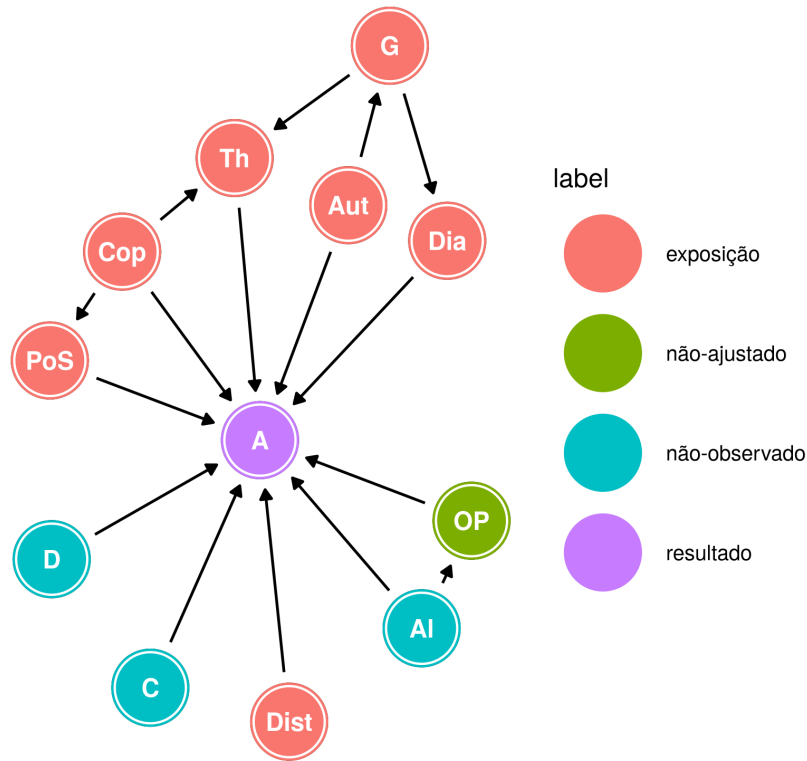


Figura 4: DAG with Attraction, **Domain**, **Controller**, **Target**, **Part of Speech**, **Copula**, **Thematic** role, **Authorship**, **Genre**, **Dialect**, **Distance**, and **Word Order**.

Modelo causal final:



Modelo linear generalizado:

$$\begin{aligned}
 A &\sim \text{Bernoulli}(p_i) \\
 \text{logit}(p_i) &= \alpha_{\text{Sent}} + \beta_{\text{Cop}} \text{Cop}_i + \beta_{\text{Dist}} |\text{Dist}_i| + \\
 &\quad + \beta_{\text{PoS}} \sum_{j=0}^{\text{PoS}_i-1} \delta_{j_{\text{PoS}}} \\
 &\quad + \beta_{\text{Th}} \sum_{j=0}^{\text{Th}_i-1} \delta_{j_{\text{Th}}} \\
 &\quad + \beta_{\text{Auth}_i} + \beta_{\text{Dia}_i} \\
 \alpha &= \bar{\alpha}_{\text{Sent}} + z_{\text{Sent}} * \sigma_{\text{Sent}} \\
 \beta_{\text{Cop}}, \beta_{\text{Dist}} &\sim \text{Normal}(0, 1) \\
 \beta_{\text{PoS}}, \beta_{\text{Th}} &\sim \text{Normal}(0, 1) \\
 \delta_{\text{PoS}}, \delta_{\text{Th}} &\sim \text{Dirichlet}(2) \\
 \beta_{\text{Auth}} &= z_{\text{Auth}} * \sigma_{\text{Auth}} \\
 \beta_{\text{Dia}} &= \bar{\beta}_{\text{Dia}} + z_{\text{Dia}} * \sigma_{\text{Dia}} \\
 \bar{\alpha}_{\text{Sent}}, \bar{\beta}_{\text{Dia}}, z_{\text{Sent}}, z_{\text{Auth}}, z_{\text{Dia}} &\sim \text{Normal}(0, 1) \\
 \sigma_{\text{Sent}}, \sigma_{\text{Auth}}, \sigma_{\text{Dia}} &\sim \text{Exponencial}(1)
 \end{aligned}$$

Cópula e distância

Effect	mean	sd	5.5%	94.5%	rhat
β_{Copula}	1.81	0.47	1.07	2.57	1.00
β_{Dist}	-0.27	0.06	-0.37	-0.18	1.01

Classe de palavra e hierarquia de predicado

Effect	mean	sd	5.5%	94.5%	rhat
β_{PoS}	0.76	0.51	0.17	1.71	1
δ_{Adj}	0.48	0.18	0.19	0.78	1
δ_{Noun}	0.52	0.18	0.22	0.81	1

PoS	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Participle	$\beta_{\text{PoS}} * 0$	0.00	0.00	0.00	0.00	–
Adjective	$\beta_{\text{PoS}} * \delta_{\text{Adj}}$	0.35	0.26	0.06	0.86	1
Noun	$\beta_{\text{PoS}} * (\delta_{\text{Adj}} + \delta_{\text{Noun}})$	0.76	0.51	0.17	1.71	1

Papel temático do controlador

Effect	mean	sd	5.5%	94.5%	rhat
β_{Theta}	2.73	0.60	1.88	3.76	1
δ_{Exp}	0.38	0.12	0.19	0.57	1
δ_{Agent}	0.62	0.12	0.43	0.81	1

Theta	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Recipient	$\beta_{\text{Th}} * 0$	0.00	0.00	0.00	0.00	–
Experiencer	$\beta_{\text{Th}} * \delta_{\text{Exp}}$	1.04	0.41	0.44	1.73	1
Agent	$\beta_{\text{Th}} * (\delta_{\text{Exp}} + \delta_{\text{Agent}})$	2.73	0.60	1.88	3.76	1

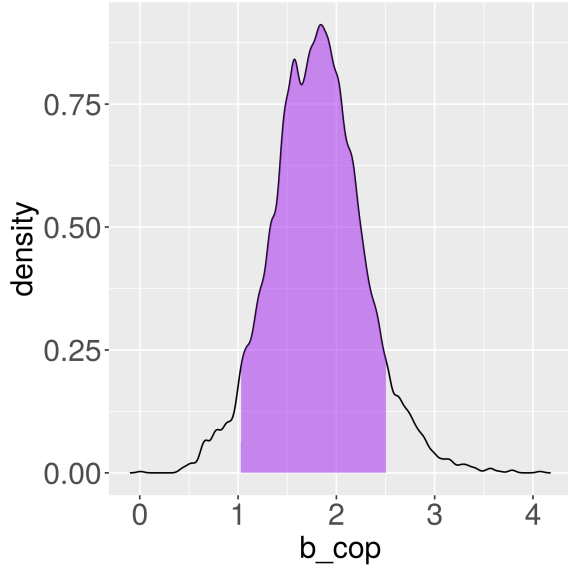


Figura 5: Distribuição posterior de β_{cop} com o 89% do Intervalo de Maior Densidade Posterior.

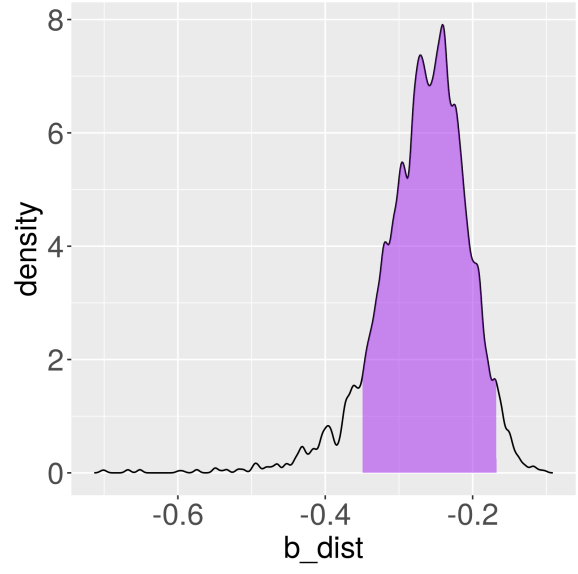


Figura 6: Distribuição posterior de β_{dist} com o 89% do Intervalo de Maior Densidade Posterior.

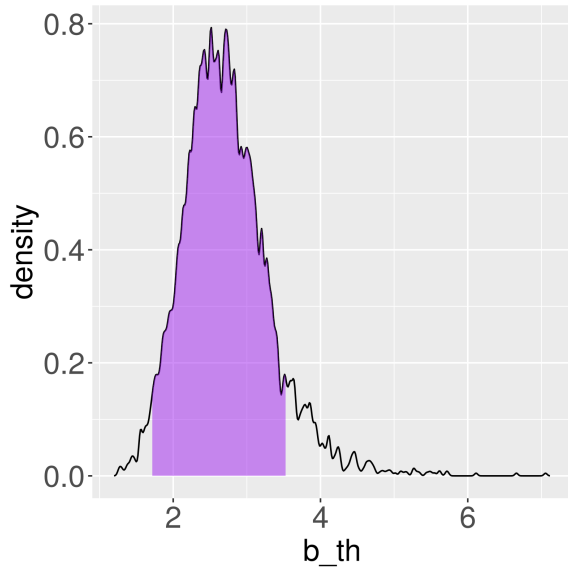


Figura 7: Distribuição posterior de β_{th} com o 89% do Intervalo de Maior Densidade Posterior.

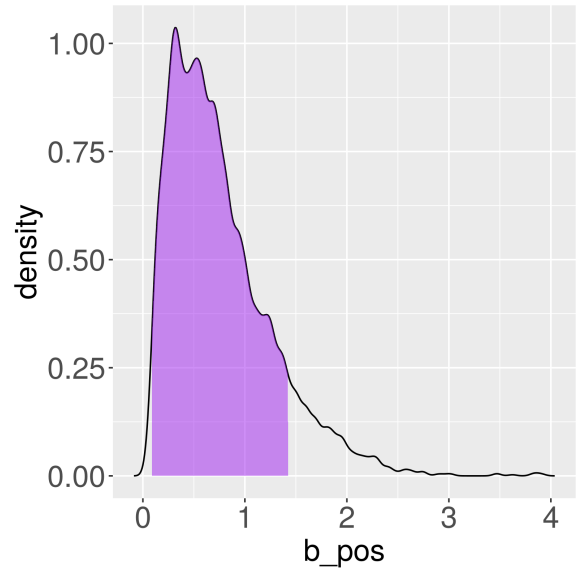


Figura 8: Distribuição posterior de β_{pos} com o 89% do Intervalo de Maior Densidade Posterior.

Dialeto e Autoria

Effect	mean	sd	5.5%	94.5%	rhat
$\bar{\beta}_{\text{Dia}}$	-0.37	0.80	-1.64	0.93	1
σ_{Dia}	1.33	0.86	0.27	2.90	1
β_{Attic}	0.13	0.87	-1.26	1.50	1
β_{Jonic}	-1.60	1.14	-3.50	0.17	1
$\beta_{\text{Attic}} - \beta_{\text{Jonic}}$	1.73	0.94	-0.04	3.01	–

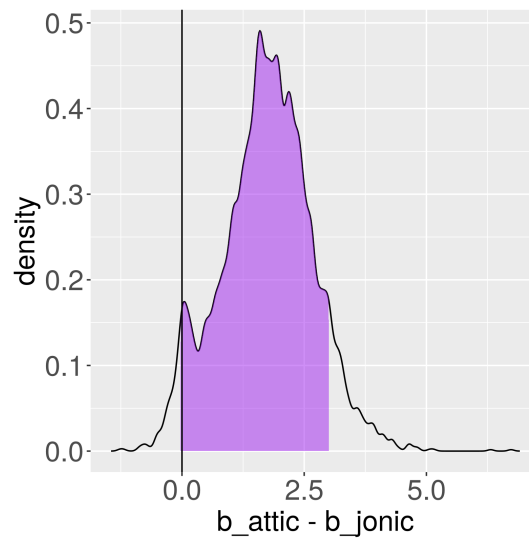


Figura 9: Distribuição posterior da diferença entre β_{Attic} e β_{Jonic} com o 89% do Intervalo de Maior Densidade Posterior.

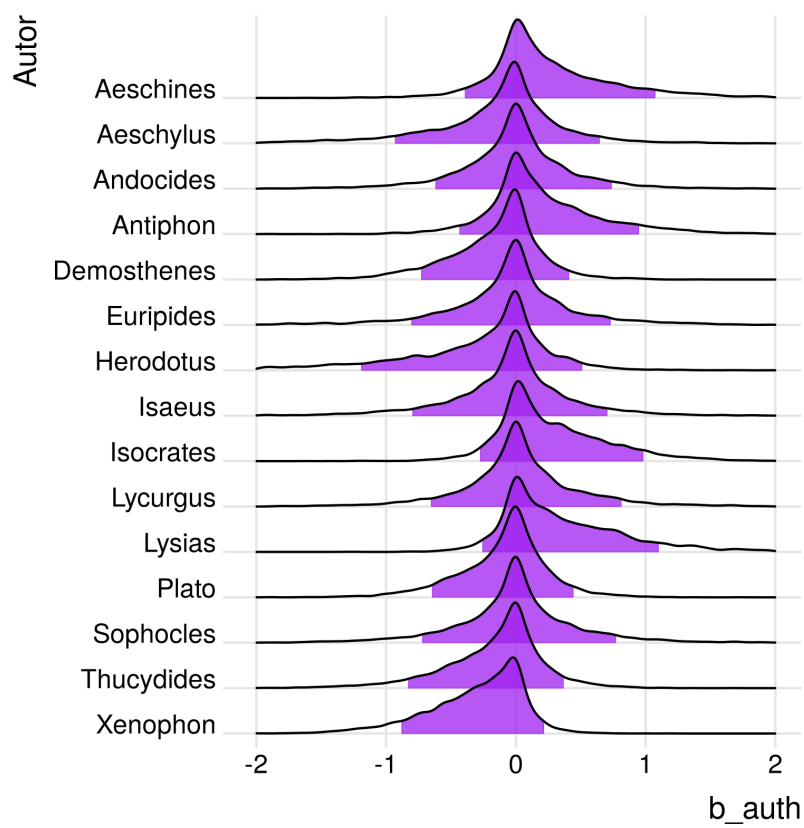


Figura 10: Distribuição posterior dos efeitos de autoria com o 89% do Intervalo de Maior Densidade Posterior.

Referências

- ANDREWS, A. D. (1971). “Case Agreement of Predicate Modifiers in Ancient Greek”. Em: *Linguistic Inquiry* 2.2, pp. 127–151.
- BARRETT, M. (2025). *ggdag: Analyze and Create Elegant Directed Acyclic Graphs*. R package version 0.2.13.9000. URL: <https://github.com/r-causal/ggdag>.
- CORBETT, G. G. (2006). *Agreement*. Cambridge: Cambridge University Press.
- GERALDES, C. B. A. (2020). “Case Attraction on Infinitive Clauses of Ancient Greek: A case study on Herodotus, Plato and Xenophon”. Diss. de mestr. São Paulo: Universidade de São Paulo. DOI: [10.11606/D.8.2020.tde-12042021-174449](https://doi.org/10.11606/D.8.2020.tde-12042021-174449).
- (2025). “Case Attraction in Infinitive Clauses: A Distributive Account”. Em: VILLA, J. de la, A. STRIANO e R. (VERANO. *Advances in Ancient Greek Linguistics: Proceedings of the 10th International Colloquium on Ancient Greek Linguistics*. Berlin: De Gruyter, pp. 351–70. DOI: [10.1515/9783111648644-020](https://doi.org/10.1515/9783111648644-020).
- HUMBERT, J. (1945). *Syntaxe grecque*. Paris: C. Klincksieck.
- KÜHNER, R. e B. GERTH (1898). *Ausführliche Grammatik der griechischen Sprache*. Hannover: Hahn.
- LAKOFF, G. (1970). “Global rules”. Em: *Language* 46.3, pp. 627–639.
- LANCELOT, C. (1658). *Nouvelle Methode pour apprendre facilement la Langue Greque: contenant les regles de Déclinaisons, des Conjugaisons, de l' Investigation du Theme, de la Syntaxe, de la Quantité, des Accens, des Dialectes, & des Licences Poétiques*. Paris: Pierre le Petit.
- (1709). *Nouvelle Methode pour apprendre facilement la Langue Latine: contenant les regles de Genres, des Déclinaisons, des Préterits, de la Syntaxe, de la Quantité, & des Accens*. Paris: Denys Marietti.
- MCELREATH, R. (2020). *Statistical Rethinking*. Boca Raton: Taylor & Francis Group, LLC.
- QUICOLI, A. C. (1982). *The Structure of Complementation*. Ghent: E. Story-Scientia.
- R CORE TEAM (2013). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria. URL: <http://www.R-project.org/>.
- SEVDALI, C. (2007). “Infinitival clauses in Ancient Greek: overt and null subjects, the role of case and focus”. Tese de dout. Cambridge: University of Cambridge.
- (2013). “Case Transmission beyond Control and the Role of Person”. Em: *Journal of Historical Syntax* 2.4, pp. 1–52.
- SPYROPOULOS, V. (2005). “The syntax of Classical Greek infinitive”. Em: *Universal Grammar in the Reconstruction of Ancient Languages*. Ed. por K. KISS. Berlin: Mouton de Gruyter, pp. 295–337.
- STAN DEVELOPMENT TEAM (2025a). *Stan Reference Manual*. Versão v2.37.0-rc2. URL: <https://mc-stan.org>.
- (2025b). *Stan User's Guide*. Versão v2.37.0-rc2. URL: <https://mc-stan.org>.
- TANTALOU, N. (2003). “Infinitives with overt subjects in Classical Greek”. Em: *Μελέτες για την ελληνική γλώσσα: πρακτικά της 23ης ετήσιας συνάντησης του τμήματος φιλολογίας του Αριστοτελείου Πανεπιστημίου Θεσσαλονίκης, 17-19 Μαΐου, 2002*. Θεσσαλονίκη: Αφοί Κυριακίδη, pp. 358–365.
- TEXTOR, J. et al. (2016). “Robust causal inference using directed acyclic graphs: the R package ‘dagitty’”. Em: *International Journal of Epidemiology* 45, pp. 1887–1894.
- VATRI, A. (2020-2022). *Diorisis Search* 3.11. Versão 3.11. URL: <https://www.crs.rm.it/diorisissearch/> (acesso em 28/02/2023).

- VATRI, A. e B. MCGILLIVRAY (2018). “The Diorisis Ancient Greek Corpus”. Em: *Research Data Journal for the Humanities and Social Sciences* 3.1, pp. 55–65. URL: https://brill.com/view/journals/rdj/3/1/article-p55_55.xml.
- (2020). “ Lemmatization for Ancient Greek: An experimental assessment of the state of the art ”. Em: *Journal of Greek Linguistics* 20.2, pp. 179–196. DOI: <https://doi.org/10.1163/15699846-02002001>.
- WICKHAM, H. et al. (2019). “Welcome to the tidyverse”. Em: *Journal of Open Source Software* 4.43, p. 1686. DOI: [10.21105/joss.01686](https://doi.org/10.21105/joss.01686).